

Data-Driven Modeling of Digital Twins: Challenges and Opportunities

Paulo Victor Lopes^{1*}, Giovanni Lugaresi² and Filipe A. N. Verri¹

¹ Department of Computer Science, Instituto Tecnológico de Aeronáutica, 12228-900 SP, Brazil

² Department of Mechanical Engineering, KU Leuven, Celestijnenlaan 300, 3001 Leuven, Belgium

* Corresponding author: paulo.lobes@ga.ita.br

Short Abstract

Digital Twins (DTs) provide a suitable solution for manufacturing systems analysis by simulating disruptive scenarios for better decision-making, though their complexity and constant updates remain challenging. Data-Driven Modeling of Digital Twins (D3T) emerges as a promising approach to automate the discovery and updating of these models. This work addresses three key gaps in D3T implementation: data collection, model discovery/validation, and what-if scenario analysis. Finally, proposing the use of artificial data generation, process mining, and automatic simulation techniques to benefit manufacturing systems.

Keywords

Digital twin; digital model; simulation; data driven; process mining; artificial data generation.

Introduction

The Digital Twins enable simulations of what-if scenarios to support more accurate decision-making. However, their increasing complexity and constant changes in real-world systems make the modeling phase a time-consuming task. The Data Driven Modeling of Digital Twins (D3T) techniques arise as one possible solution for this challenge (Friederich et al., 2022).

Contribution

The D3T techniques still in early stages compared to the DT concept constraining the development of new tools. Thus, this work aims to contribute for DT researchers to understand the D3T field introducing properly the general D3T framework, with practical examples, and to provide a description of future opportunities. In order to motivate this description, it is possible to highlight three major implementation gaps for the D3T framework: i) data collection and benchmark (Lopes et al., 2024), ii) discovery and validation of simulation models (Lugaresi and Mata, 2021) and iii) what-if scenarios analysis (Kumbhar et al., 2023).

Conclusions

The three main implementation gaps can be overcome by different approaches, but the following are highlighted: the use of artificial data generation strategies, process mining tools, automatic simulation model generation techniques, online validation and the manufacturing systems performance evaluation strategies. Finally, the development of demonstrators is valuable to provide better examples and standards for practitioners.

References

- Friederich, J., Francis, D. P., Lazarova-Molnar, S., & Mohamed, N. (2022). A framework for data-driven digital twins of smart manufacturing systems. *Computers in Industry*, 136, 103586.
- Lopes, P. V., Silveira, L., Guimaraes Aquino, R. D., Ribeiro, C. H., Skoogh, A., & Verri, F. A. N. (2024). Synthetic data generation for digital twins: enabling production systems analysis in the absence of data. *International Journal of Computer Integrated Manufacturing*, 1-18.
- Lugaresi, G., & Matta, A. (2021). Automated manufacturing system discovery and digital twin generation. *Journal of Manufacturing Systems*, 59, 51-66.
- Kumbhar, M., Ng, A. H., & Bandaru, S. (2023). A digital twin based framework for detection, diagnosis, and improvement of throughput bottlenecks. *Journal of manufacturing systems*, 66, 92-106.
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